Department of Computer Science And Engineering (CSE)
BRAC University

CSE 251 – Electronic Devices and Circuits

LECTURE: ##

MOSFET

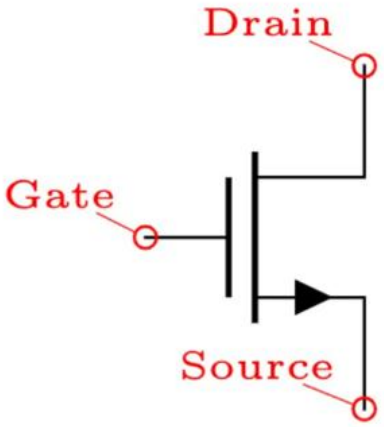
PREPARED BY
Mir Hamidul Hussain (HMH)
Lecturer
Department of CSE, BRAC University
Email: hamidul.hussain@bracu.ac.bd

OUTLINE

- Construction of Real MOSFET
- Device Structure of NMOS
- Modes of Operation
- NMOS I-V Characteristics
- Real MOSFET Equations
- Solving Circuits with MOSFET

Introduction to Real MOSFET

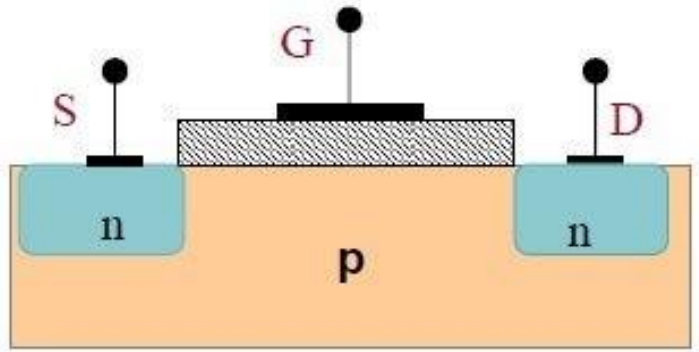
- **M**etal **O**xide **S**emiconductor **F**ield **E**ffect **T**ransistor



Symbol



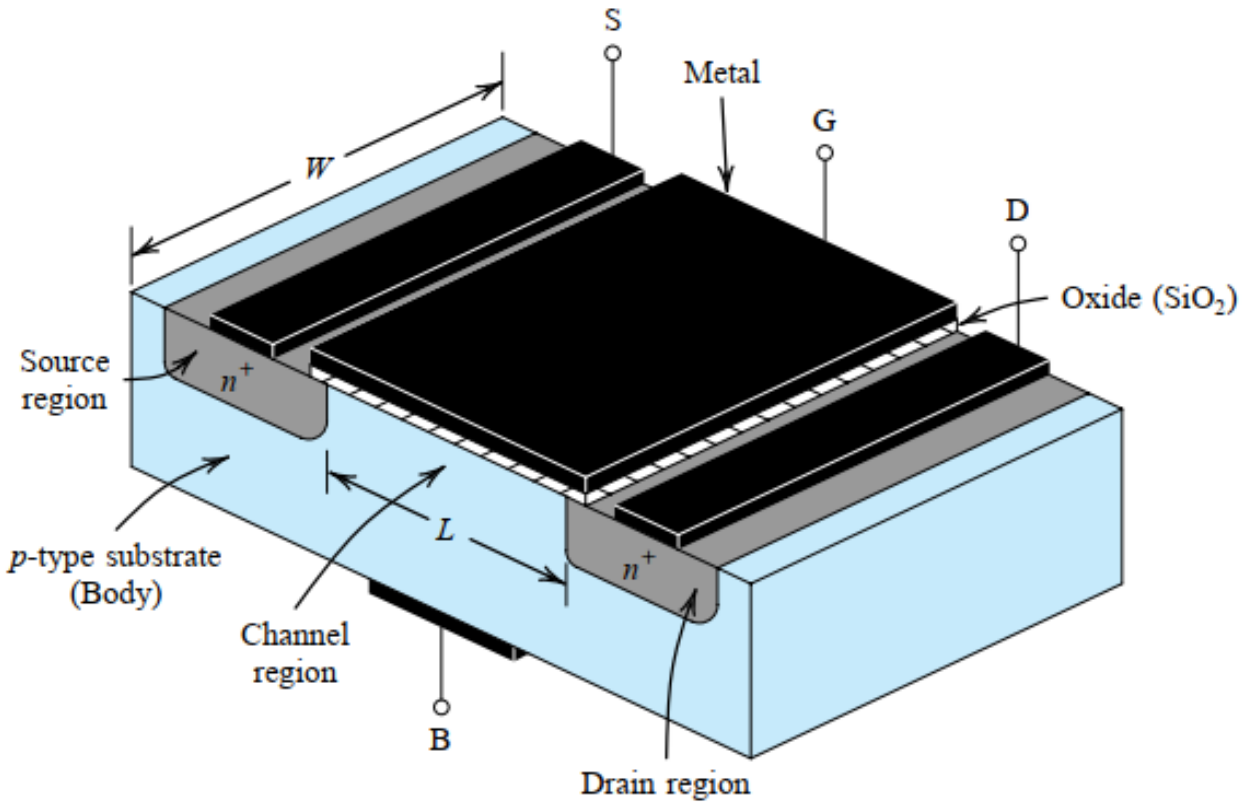
Real MOSFET



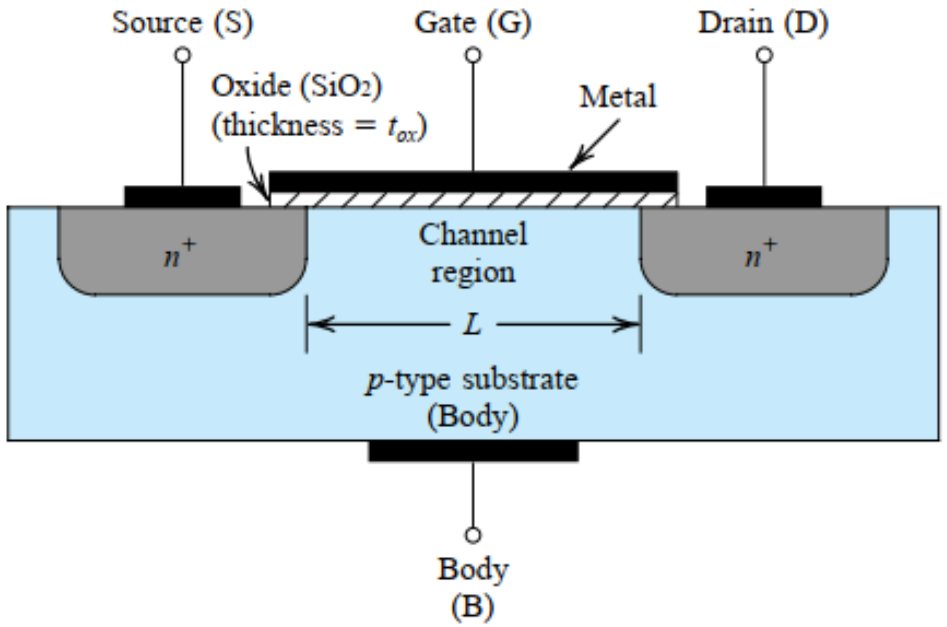
Physical Structure

Construction of Real MOSFET

Physical Structure of N-channel Enhancement Type MOSFET



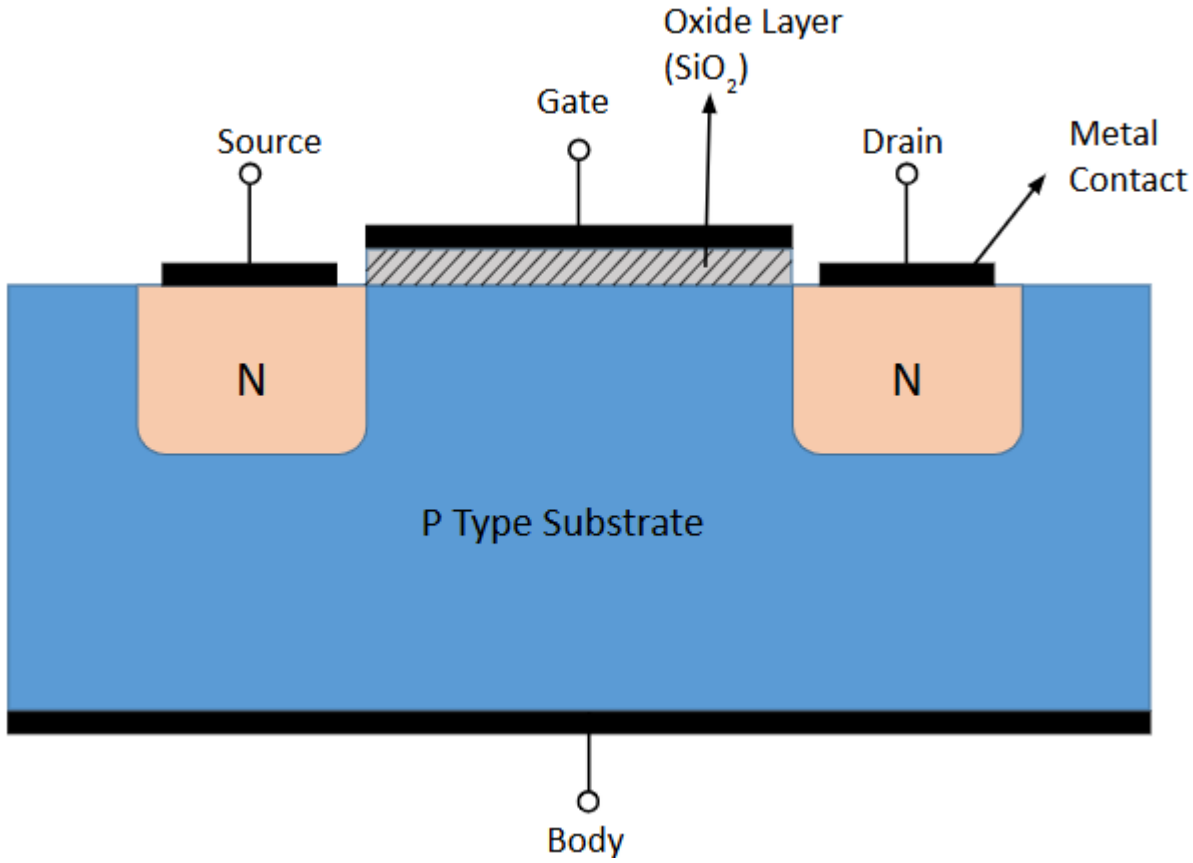
perspective view



cross-sectional view

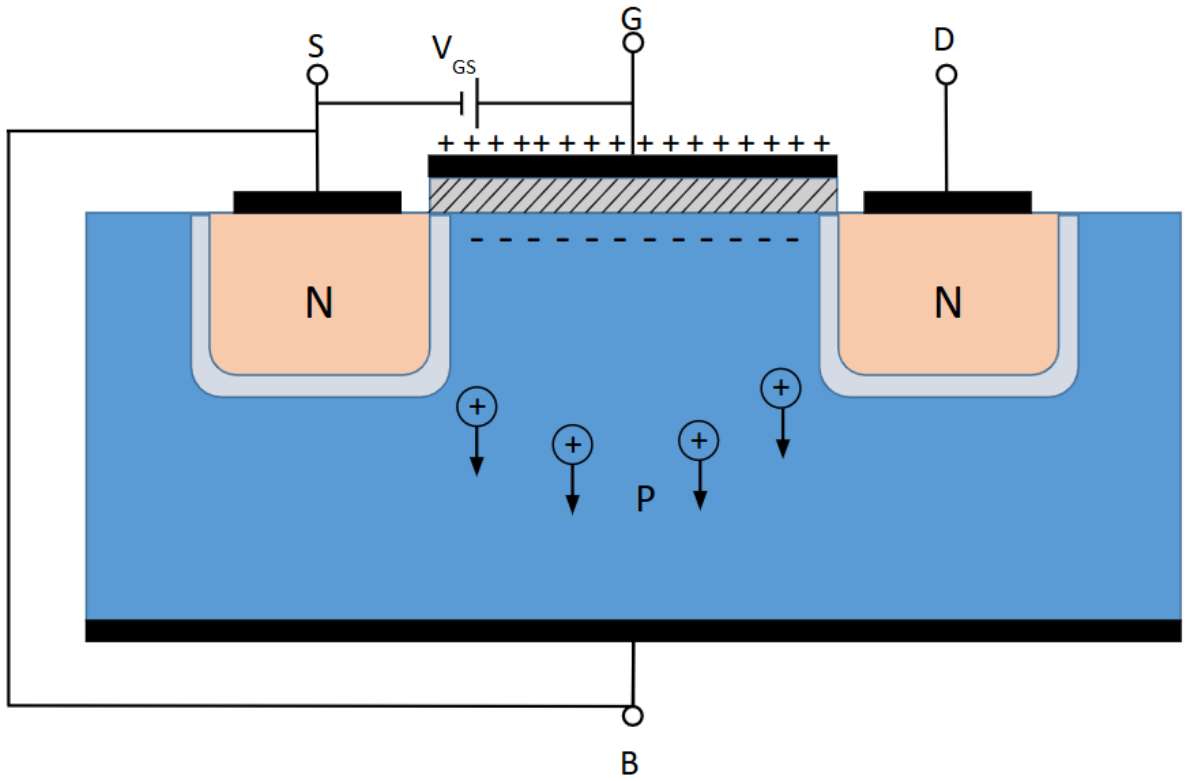
Device Structure of NMOS

- The transistor is fabricated on a P-type substrate.
- Two heavily doped N-type regions are formed in the substrate- known as the **source** and other one is the **drain**.
- A thin layer of silicon dioxide (SiO_2), which is an insulator, is grown on the surface of the substrate, covering the area between the source and the drain.
- Metal is deposited on the top of the oxide layer to form **gate electrode**.
- Metal contact are also made on the source region, the drain region and the substrate (**body**).

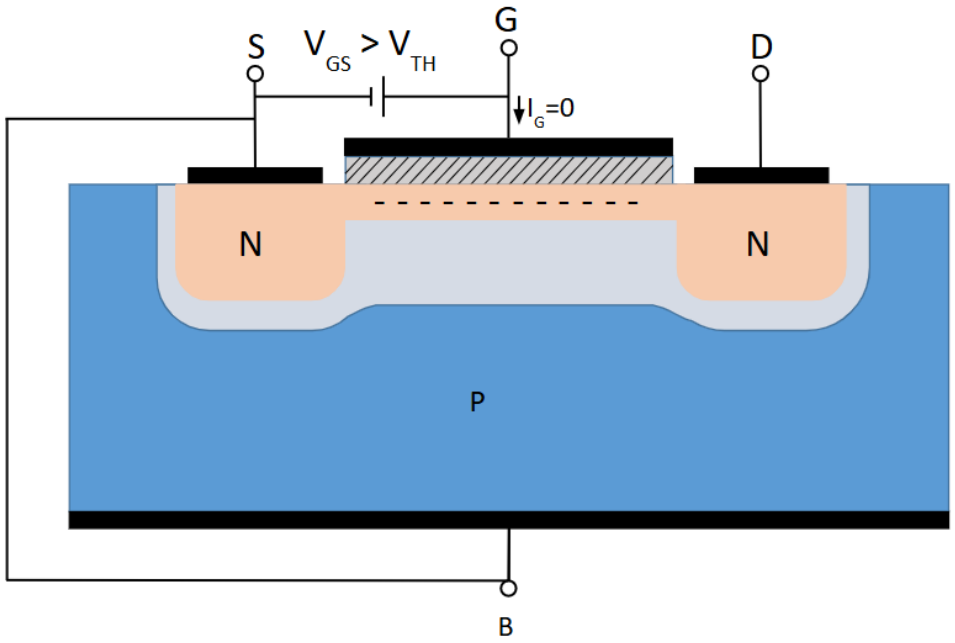


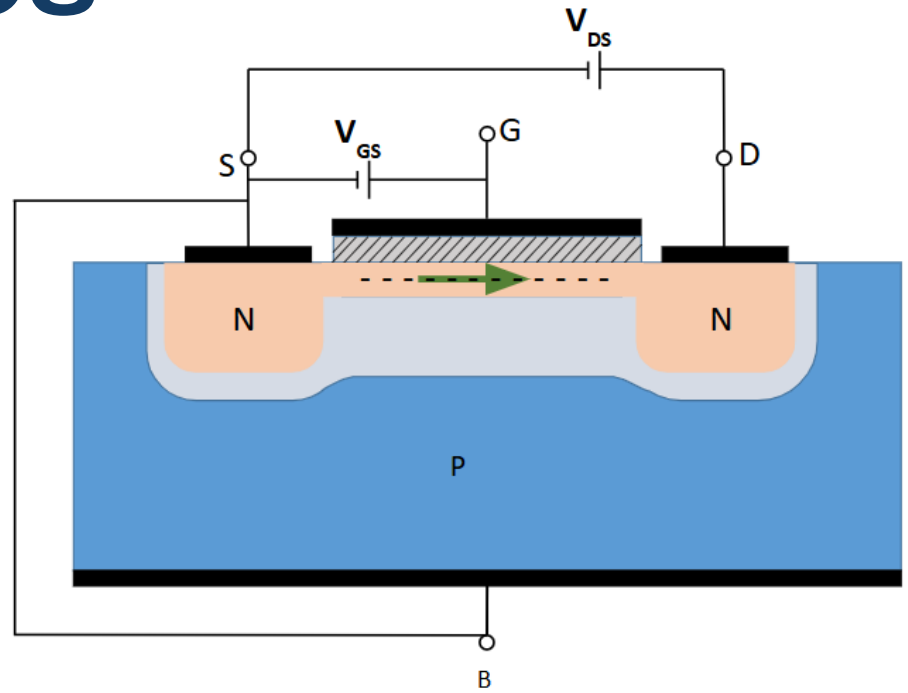
Device Structure of NMOS

- The substrate forms PN junctions with the source and the drain regions
- When a small positive V_{GS} is applied, the free holes are repelled from the region of the substrate under the gate.
- These holes are pushed downward into the substrate, leaving behind a carrier-depletion region.
- This depletion region is populated by the bound negative charge associated with the acceptor atoms



Device Structure of NMOS

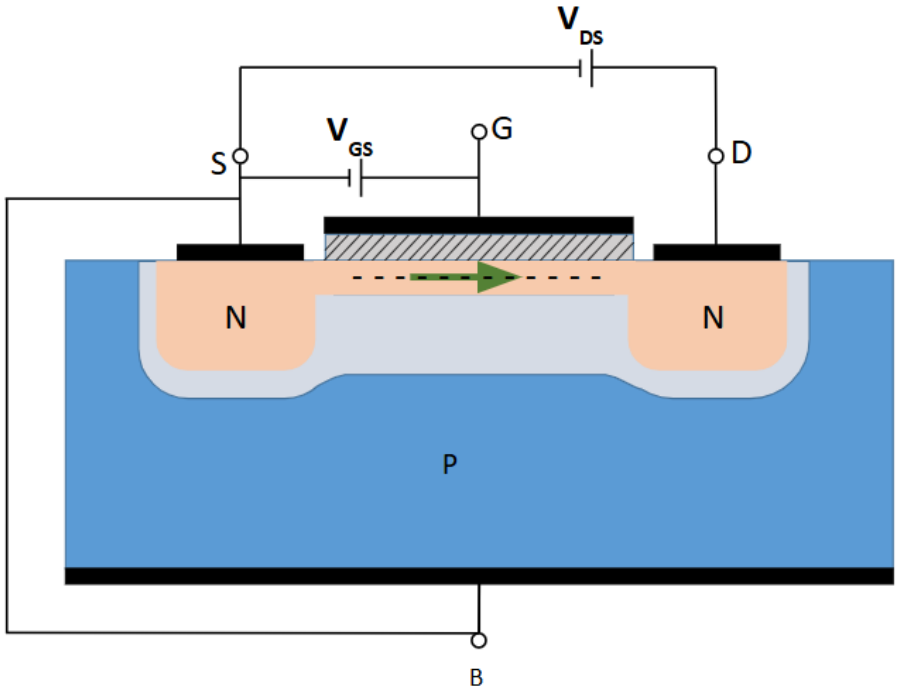


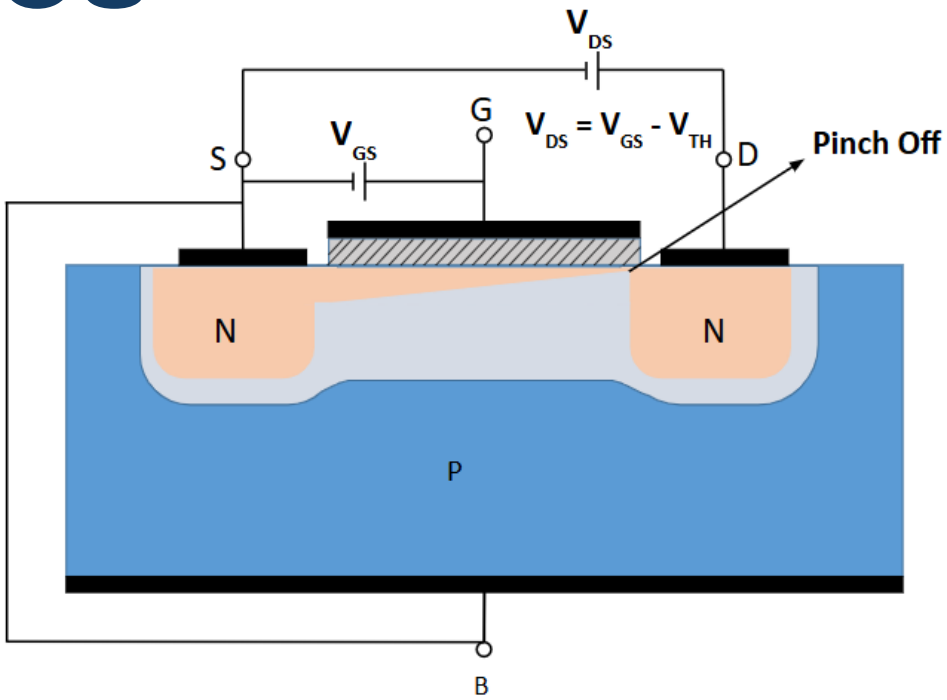


- When V_{GS} is larger than a threshold voltage V_{TH}, a sufficient number of electrons accumulate near the surface of the substrate under the gate and creates an induced n-type channel.

- Now if a voltage (V_{DS}) is applied between drain and source, the free electrons flow from source to drain . (positive current flows from drain to source)

Device Structure of NMOS





- For small V_{DS} , the channel is uniform and behaves like a linear resistance whose value is controlled by the **overdrive voltage V_{OV}** . Overdrive voltage is the excess of V_{GS} over V_{TH} i.e. **$V_{OV} = V_{GS}-V_{TH}$**

- As V_{DS} is increased, the channel no longer maintain a uniform depth and becomes tapered. With further increase in V_{DS}, the channel becomes more tapered, resulting in **higher channel resistance**. When **$V_{DS} = V_{OV}$** , the channel **pinches off**

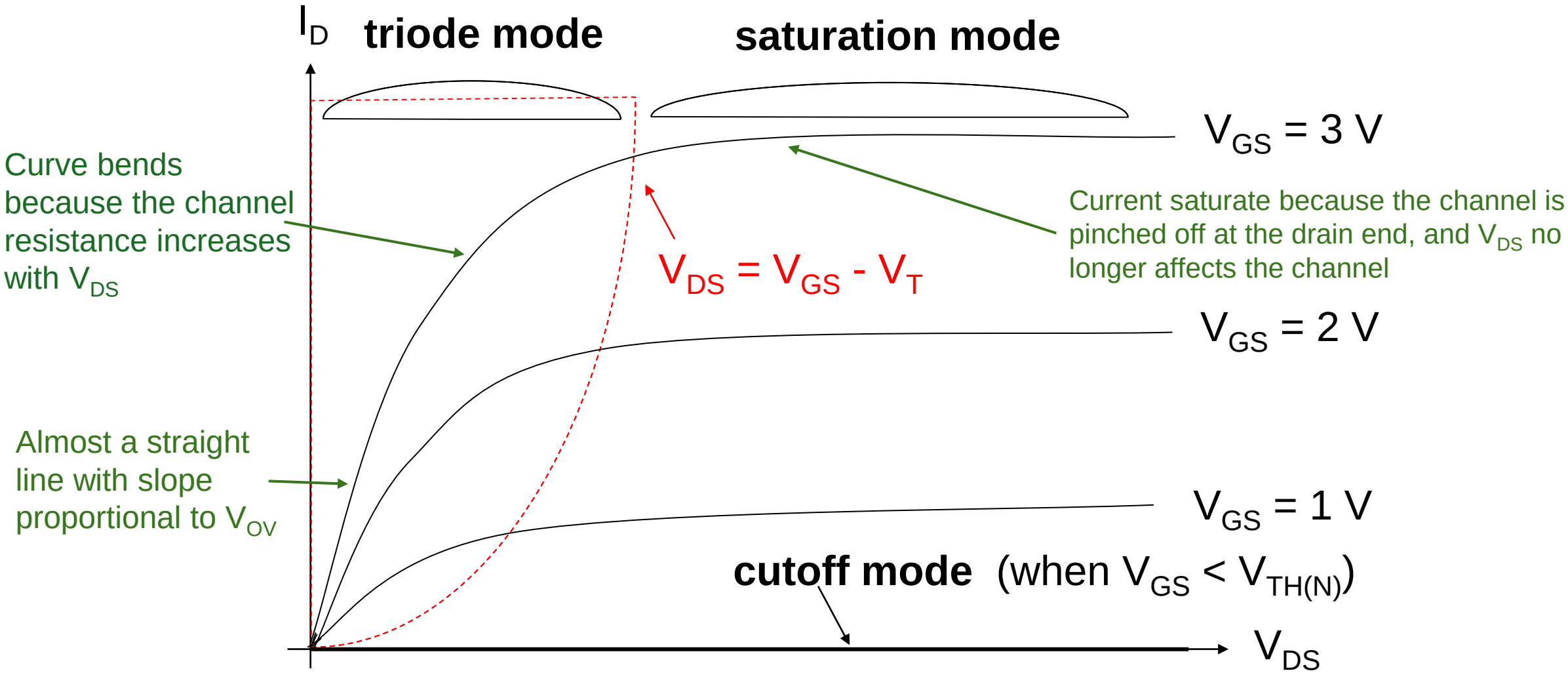
Modes of Operation

- **Cut off :** When $V_{GS} < V_{TH}$, the n-type channel is not formed. No current flows.

When $V_{GS} \geq V_{TH}$, current I_D may flow from drain to source. The flow of current depends on V_{DS} and the channel inducing voltage , $V_{GS} - V_{TH}$ (Overdrive voltage, V_{OV}) . Based on the V_{DS} , the operating modes are:

- **Triode :** When $V_{DS} < V_{OV}$, the MOSFET behaves as a linear resistance, whose value is controlled by V_{GS}
- **Saturation:** When $V_{DS} \geq V_{OV}$, the channel is pinched off and the drain current I_D become independent of the V_{DS} for a given V_{GS}.

NMOS I-V Characteristics



NMOS Equations

Mode	Conditions	Equations
Cut off	$V_{GS} < V_{TH}$	$I_D = 0$
Triode	$V_{GS} \geq V_{TH}$	$I_D = K_n[(V_{GS} - V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2]$
	$V_{DS} < V_{OV}$	
Saturation	$V_{GS} \geq V_{TH}$	$I_D = \frac{K_n}{2}(V_{GS} - V_{TH})^2$
	$V_{DS} \geq V_{OV}$	

$K_n = MOSFET\ Transconductance$

$K_n = K'_n \frac{w}{L}$

$\frac{W}{L} = Aspect\ Ratio$

$W = Channel\ Width$

$L = Channel\ Length$

$K'_n = \mu_n C_{ox}$

$\mu_n = Electron\ Mobility$

$C_{ox} = Oxide\ Capacitance$

Solving Circuits with MOSFET

- Use **Method of Assumed State**

- Three Steps:
  - **Assume**: One of the modes (cut off, triode, saturation)
  - **Solve**: Use corresponding equation and KCL+KVL or any other circuit analysis method.
  - **Verify**: Check if the conditions of $\mathbf{V_{DS}}$ and $\mathbf{V_{GS}}$ are satisfied. If not then repeat.
- Might need to solve quadratic equations $(ax^2 + bx + c = 0)$.
- If we get two roots, choose the one that is favourable to your assumption.

Example

1. Analyze the circuit to find i_D and V_D using method of assumed state. here the input of the MOSFET is $V_G = 1V$. You must validate your assumptions

Assumption:

Assume the MOSFET operates in **saturation**

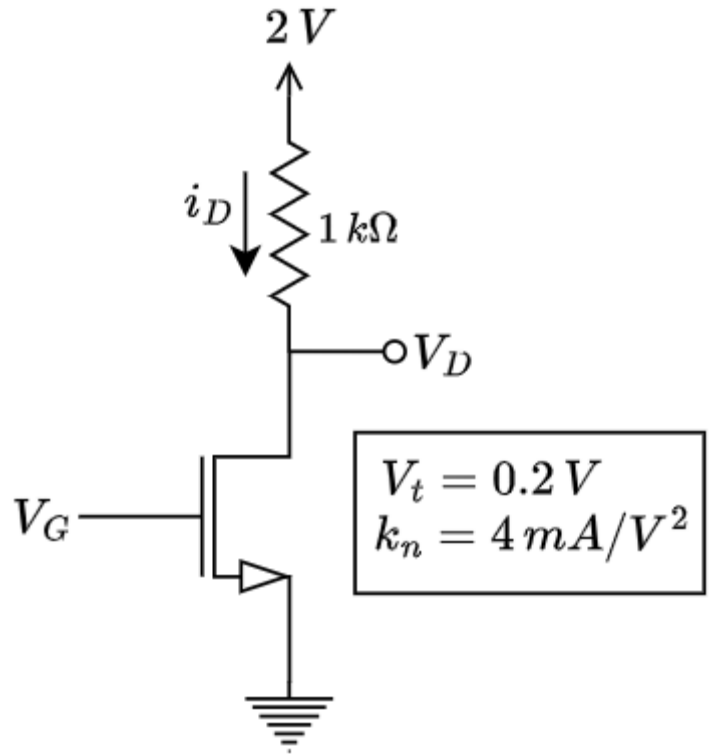
$V_{GS} \geq V_t$
$V_{DS} \geq V_{OV}$

$$V_G = 1\,V$$
$$V_{GS} = V_G - V_S = 1\,V \quad [V_S = 0\,V]$$
$$V_{OV} = V_{GS} - V_t = 0.8\,V$$

Current equation, $i_D = \frac{K_n}{2}(V_{GS} - V_T)^2 = 1.28\,mA$

$$i_D = \frac{2 - V_D}{1\,k\Omega}$$

$$V_D = 2 - i_D = 0.72\,V$$



Verification:

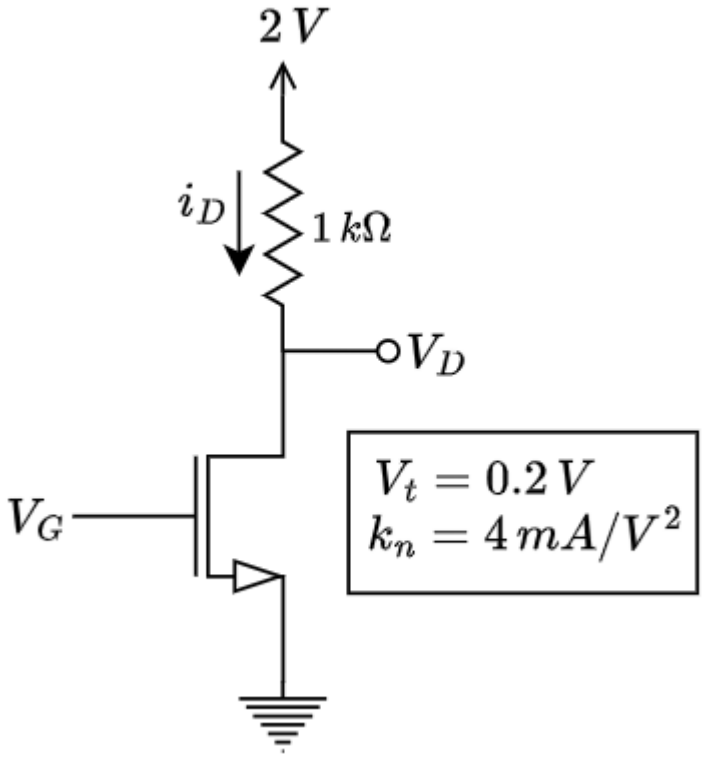
For Saturation mode: $V_{DS} \geq 0.8\,V$
But here, $V_{DS} = 0.72V < 0.8\,V$
So the conditions are not satisfied. → Assumption is incorrect

Example

1. Analyze the circuit to find i_D and V_D using method of assumed state. here the input of the MOSFET is $V_G = 1V$. You must validate your assumptions

Assume the MOSFET is in **Triode** mode.

$V_{GS} \geq V_t$
$V_{DS} < V_{OV}$



$$i_D = K_n(V_{GS} - V_t - \tfrac{1}{2}V_{DS})V_{DS} = \tfrac{2-V_D}{1}$$
$$\Rightarrow K_n(V_G - V_t - \tfrac{1}{2}V_D)V_D = 2 - V_D$$
$$\Rightarrow 4(0.8 - \tfrac{1}{2}V_D)V_D = 2 - V_D$$
$$\Rightarrow V_D = \cancel{1.37\,V}\ or\ 0.73\,V$$

$V_{DS} = 0.73\,V$
$V_{OV} = 0.8\,V$

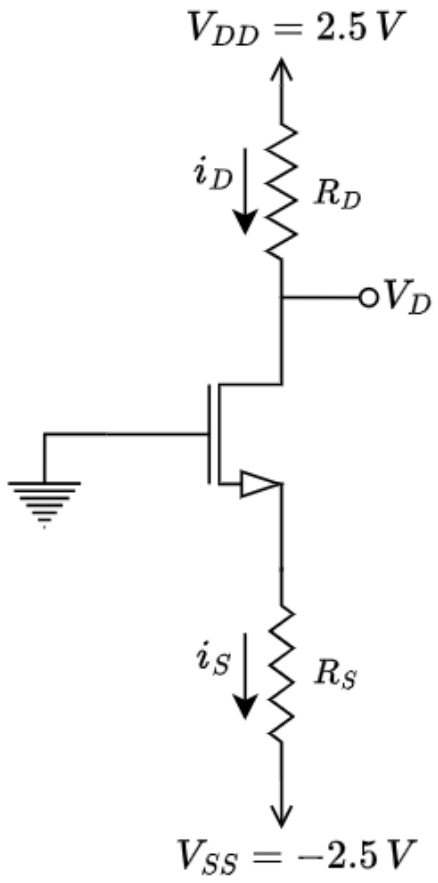
$V_{DS} < V_{OV} \rightarrow$ Assumption is correct

$i_D = \tfrac{2-V_D}{1} = 1.27\,mA$

Example

2. Determine the values of R_D and R_S , so that the transistor operates at $i_D = 0.4\,mA$ and $V_D = +0.5\,V$. The NMOS transistor has $V_t = 0.7\,V$, $K_n = 3.2\,mA/V^2$.

$$R_D = \frac{V_{DD} - V_D}{i_D} = \frac{2.5 - 0.5}{0.5} = 5\,k\Omega$$



Assumption:

Assume the **NMOS** is operating in the **Saturation** region.

$V_{GS} \geq V_t$

$V_{DS} \geq V_{OV}$

$V_{GS} = -V_S \quad [V_G = 0\,V]$

$V_{OV} = V_{GS} - V_t = -V_s - 0.7$

Current equation, $i_D = \frac{K_n}{2}(V_{GS} - V_t)^2 = \frac{K_n}{2}(-V_S - V_t)^2$

$\Rightarrow 0.4 = \frac{3.2}{2}(-V_S - 0.7)^2$

$\Rightarrow V_S = \cancel{-0.2\,V}$ *or* $-1.2\,V$

Example

2. Determine the values of R_D and R_S , so that the transistor operates at $i_D = 0.4\,mA$ and $V_D = +0.5\,V$. The NMOS transistor has $V_t = 0.7\,V$, $K_n = 3.2\,mA/V^2$.

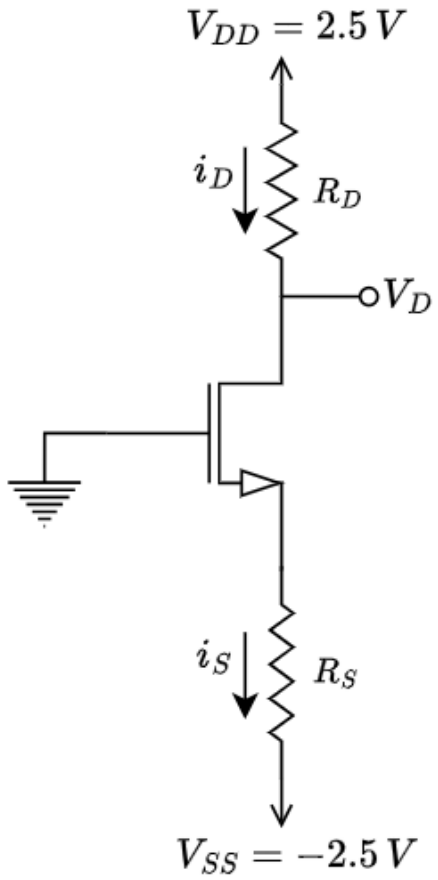
Verification:

$$V_{DS} = 0.5 - V_S = 1.7\,V$$

$$V_{OV} = V_{GS} - V_t = -V_S - 0.7 = 0.5\,V$$

$$V_{DS} \geq V_{OV} \rightarrow \text{Assumption is correct}$$

$$R_S = \frac{V_S - V_{SS}}{i_D} = \frac{-1.2-(-2.5)}{0.4} = 5.75\,k\Omega$$



Example

3. Analyze the circuit to determine V_D, V_S and I_D. The NMOS transistor has $V_t = 1\,V$, $K_n = 1\,mA/V^2$.

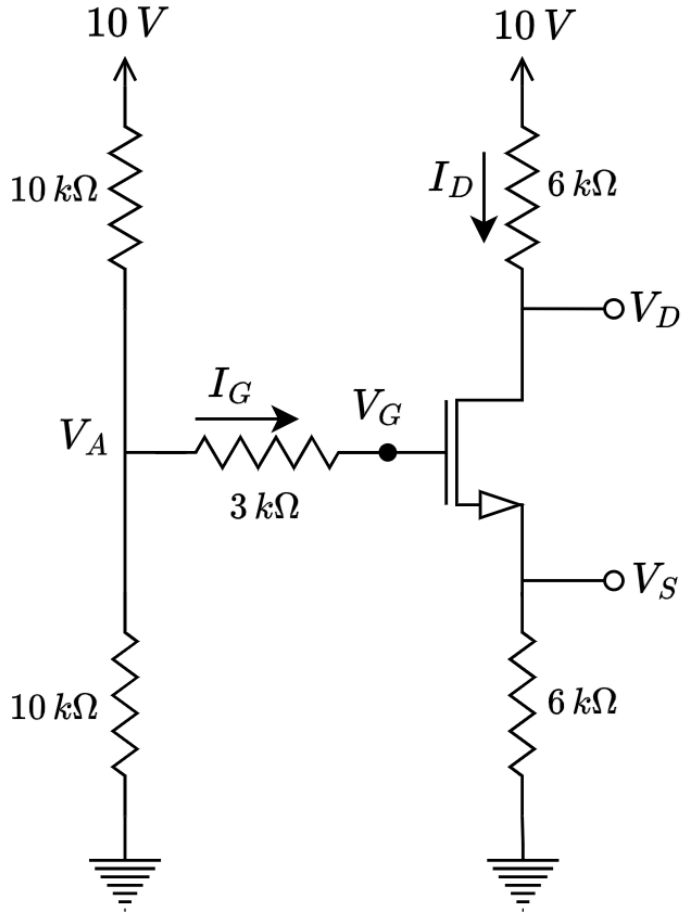
Since the gate current I_G of a MOSFET is zero $(i.e., I_G = 0)$, the Voltage V_A can be determined using nodal analysis or the voltage division rule.

$$\frac{V_A - 10}{10} + \frac{V_A}{10} = 0$$

$$\Rightarrow V_A = 5\,V$$

$$V_G = V_A = 5\,V \;;\; [I_G = 0]$$

Now the circuit can be redrawn by labeling the gate terminal with the voltage $V_G = 5\,V$.



Example

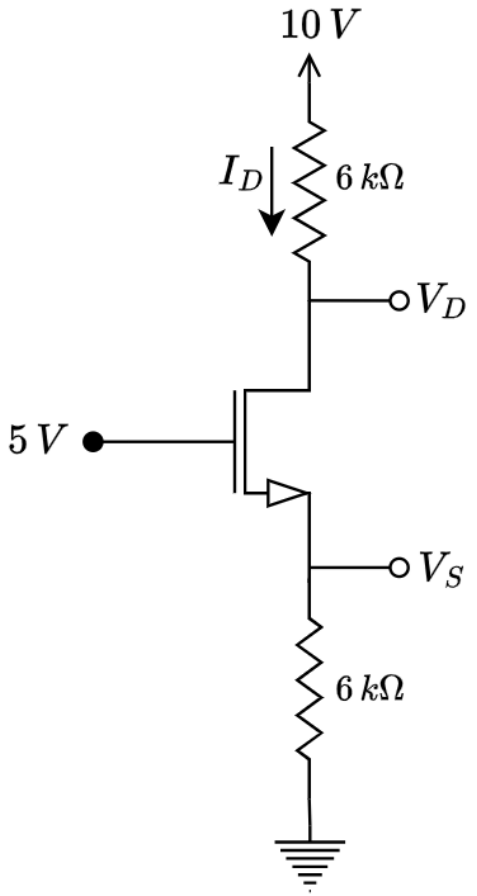
3. Analyze the circuit to determine V_D, V_S and I_D. The NMOS transistor has $V_t = 1\,V$, $K_n = 1\,mA/V^2$.

Assumption:

Assume the MOSFET is in **saturation** mode.

$$V_{GS} \geq V_t$$
$$V_{DS} \geq V_{OV}$$

$$I_D = \frac{10 - V_D}{6} = \frac{V_S}{6}$$



Current equation,

$$i_D = \frac{K_n}{2}(V_{GS} - V_t)^2 = \frac{K_n}{2}(5 - V_S - 1)^2$$

$$\Rightarrow \frac{V_S}{6} = \frac{1}{2}(4 - V_S)^2$$

$$\Rightarrow V_S = \cancel{5.33\,V}\ or\ 3\,V$$

Example

3. Analyze the circuit to determine V_D, V_S and I_D. The NMOS transistor has $V_t = 1\,V$, $K_n = 1\,mA/V^2$.

Verification:

Drain current, $i_D = \frac{V_S}{6} = 0.5\,mA$

Drain Voltage, $V_D = 10 - 6i_D = 7\,V$

$V_{DS} = V_D - V_S = 7 - 3 = 4\,V$

$V_{GS} - V_t = V_G - VS - V_t = 1\,V$

For saturation mode,

$V_{DS} \geq V_{GS} - V_t \rightarrow$ Assumption is correct

